STRESS ZOMBIE APOCALYPSE (v4.7)

Interactive 2D Stress/Strain Transformation, Mohr's Circle & Failure Theory Simulation Engine

Course Alignment: Mechanics of Solids (Plane Stress & Strain, Mohr's Circle 2θ, Generalized Hooke's Law, Strain Energy & Failure Criteria)
Runtime Stack: HTML5 Canvas (5 Canvases) + Vanilla ES6 JavaScript (6 Modules) + Procedural Web Audio + Python 3 HTTP Server
Codebase & Verification: 12,047 source lines across 10 files • 11/11 analytical unit tests passing • Zero external dependencies

1. What the Game Is About: Executive Summary & Problem Statement

Stress Zombie Apocalypse is a browser-based serious engineering game designed to turn abstract 2D continuum mechanics equations into real-time visual and tactile mastery. In the game scenario, biomechanically altered zombies advance across a quarantined horror highway toward the player's defense barricade. Each enemy is structurally bound by an internal **2D Cauchy plane stress tensor** (normal stresses σ_x, σ_y and shear stress τ_{xy} in MPa), displayed continuously via a live floating stress cube over its head.

Unlike conventional arcade shooters where random firing works, **conventional gunfire cannot hurt any enemy off-axis**. The player commands a barricade-mounted **7.62mm Heavy Machine Gun** equipped with a rotating breech handling crank (angle **θ** from 0° to 180°). Its safety interlock physically prevents gunfire unless the player aligns θ with the enemy's true material failure plane—either the **Rankine maximum principal stress plane (θ_p)**, the **Tresca maximum in-plane shear plane ($\theta_s = \theta_p \pm 45°$)**, or the **von Mises distortion-energy plane**.

Key Pedagogical Breakthrough — Overcoming the 2θ Confusion:
First-time Mechanics of Solids students consistently struggle to connect a physical cut rotation **θ** with its doubled angle **2θ** on Mohr's Circle. Here, rotating the gun breech crank by **θ** sweeps the live transformed stress vector **$P(\sigma_\theta, \tau_\theta)$** clockwise around Mohr's Circle by exactly **2θ** (since stress equations depend on cos 2θ and sin 2θ). Players can even **click or drag directly anywhere on Mohr's Circle**, and the engine inversely solves the exact physical crank angle θ in real time.

2. Core Educational Features & Classroom Value

Feature Pillar	Engineering Mechanism & How Students Learn From It
Live Breech Crank Safety Interlock	Gun only transitions from crimson **LOCKED** to green **ARMED** inside ±7.5° of the analytical failure plane. Off-angle shots cannot spray bullets randomly; HUD guides students (e.g., *TURN [D] -> 22.5°*) until bonds yield.
Guided Tutorial Before Level 1	A non-lethal 4-drill training mission teaches: (1) waiting for on-screen visibility, (2) 2θ shear plane rotation (22.5°), (3) switching from brittle σ_1 to ductile τ_{max} (67.5°), and (4) using Hydrostatic Pressure [5] to wipe $\tau = 0$.
Every-Zombie Kill Reports & Lab	Every single kill pops a 4.5s non-blocking Live Telemetry Card AND records a permanent full-tensor Stress & Strain Autopsy (Mohr C/R, Hooke's Law ε_x, ε_y, ε_z, γ_{xy}, dilatation e, strain energy u_v/u_d) inspectable interactively after every wave.
Instructor Study Lab Sandbox	Accessible anytime without combat: instructors or students adjust sliders for σ_x, σ_y, τ_{xy}, and θ, load lecture presets (or slain zombies), and project step-by-step derivations + deformed strain squares live in class.

3. Target Users & Syllabus Coverage

Audience Group	Syllabus Modules Covered	Academic Use Case
Undergraduate Engineering Students (Mech, Civil, Aero, Materials)	Plane Stress Transformation, Principal Planes, Invariants I_1/I_2, Hooke's Law, Rankine/Tresca/von Mises	Bridges textbook derivations with instant spatial cause-and-effect during guided levels and tutorial drills.
Faculty & Teaching Assistants	Mohr's Circle Construction, Hydrostatic/Deviatoric Decomposition, Strain Energy	Zero-install classroom projector demo via Chrome; inspect every zombie's live Mohr circle and deformed element.
Competitive Exam Prep (GATE ME/CE, FE, ESE)	Rapid determination of C, R, θ_p, θ_s, and fracture criteria identification	Builds rapid geometric intuition for clockwise 2θ rotations and normal vs. shear failure recognition.

4. System Architecture & Complete Screen Flow

The application separates analytical physics, canvas rendering, procedural audio, and wave state into decoupled ES6 modules. Zero external NPM packages, WebGL engines, or external image/audio files are used:

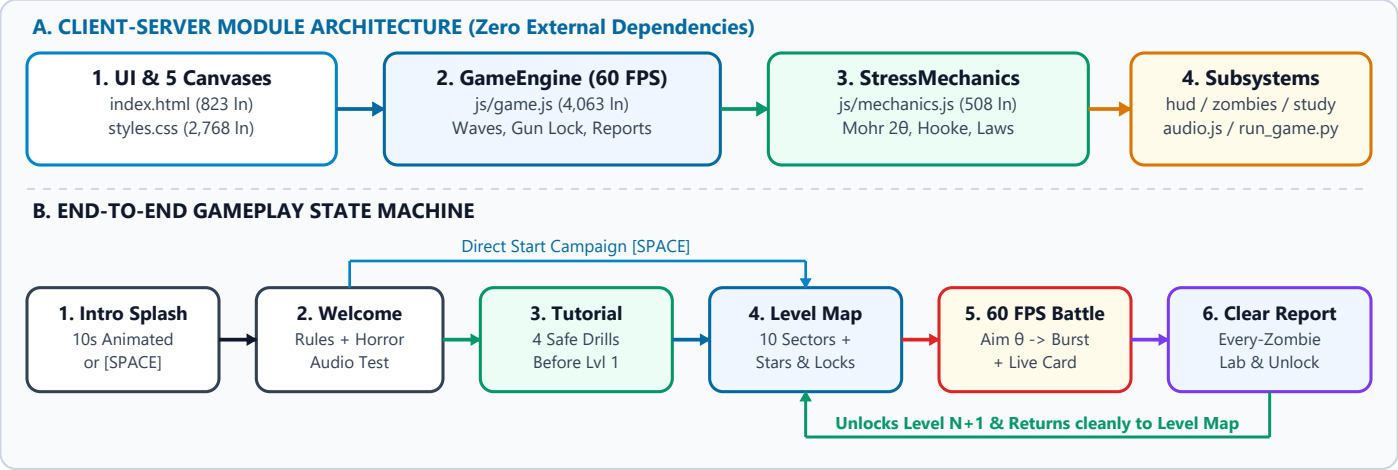


Figure 1: Modular Software Architecture (top) and User Navigation Flow from Intro Splash through Forensic Reporting (bottom).

5. Codebase File-by-File Breakdown (Total: 12,047 Lines)

File	Size / Lines	Core Symbols	Runtime Responsibility
`index.html`	823 ln / 41 KB	5 Canvases, 6 Modals	Battlefield (840×480), crank dial (130×130), Mohr radar (280×280), stress graph (280×110), intro/welcome overlays, tutorial bar, reports & Lab.
`styles.css`	2,768 ln / 64 KB	HUD & Modal Themes	Responsive tactical deck layout, horror splash lighting, floating kill card, level map cards, and energy partitioning bars.
`js/mechanics.js`	508 ln / 22 KB	`StressMechanics`	Pure analytical solver: transform, Mohr C/R/σ_1/σ_2/θ_p/θ_s, invariants, Rankine/Tresca/von Mises evaluator, Hooke strains, dilatation & autopsies.
`js/game.js`	4,063 ln / 168 KB	`GameEngine`	Master orchestrator: 10s intro splash, welcome screen, 4-drill tutorial, 10-sector wave factory, 60 FPS loop, visibility gate, bursts, horror cache & reports.
`js/hud.js`	703 ln / 27 KB	`TacticalHUD`	Syncs gun crank θ, draws rotating breech lever, renders live 2θ Mohr's Circle with pointer inverse-aiming, and plots σ_θ / $\lvert\tau_\theta\rvert$ curves (0°–180°).
`js/zombies.js`	1,119 ln / 41 KB	`Zombie`	Entity movement, hydro quench, shapeshifter 240f mutation, boss 3-phase shifts, comic-horror art, θ-plane cleavage animation & floating stress cube.
`js/study.js`	462 ln / 20 KB	`StudyLab`	Reusable forensic analyzer rendering interactive Mohr's Circle, deformed/undeformed rotated stress-strain square, slain presets & derivations.
`js/audio.js`	1,239 ln / 47 KB	`SoundController`	Procedural Web Audio synthesizer: 7.62mm bursts, trigger clicks, deflections, Hydro-Jet wash, boss alarms, 74–96 BPM BGM & 4-layer Zombie Roar.
`run_game.py`	86 ln / 2.8 KB	`CustomHandler`	Multi-threaded HTTP server (port 8000, no-cache headers) detecting LAN IP for instant Chrome play.
`test_mechanics.py`	276 ln / 11.7 KB	`TestStressMechanics`	Python unittest suite running 11 analytical checks against textbook benchmarks (invariants, strains, energies, criteria).

6. Startup & Navigation Flow (Launch -> Campaign Progression)

1. 10-Second Animated Entry Splash: On load, `initIntroSplash()` plays a full-viewport canvas showcase for `10,000 ms` featuring 5 animated zombies creeping through grave mist, cratered moon, lightning flashes (1.85s–9.05s), bats, and a faint rotating Mohr crest. Pressing **[SPACE]**, **[ENTER]**, **[ESC]**, or clicking advances immediately.
2. Streamlined Welcome Screen: Shows the 3-step rules, live 6-bar equalizer Horror Zombie Call test, Quick Guide, Study Lab, **PLAY TUTORIAL**, and **START CAMPAIGN [SPACE]**.
3. Campaign Level Map & Persistence: Stores highest unlocked level, star ratings, and high scores in browser `localStorage`. The tutorial stays accessible above Level 1; sectors unlock sequentially as waves are cleared.

7. Guided Pre-Level-1 Tutorial (4 Non-Lethal Drills)

Calling `startTutorial()` resets temporary score/streak, starts the gun crank at 45° so the student actively practices rotating θ, displays the live 4-step coaching bar (`#tutorialCoachBar`), and pauses approaching practice zombies safely at x = 270 px so the barricade is never breached:

Drill	Target Dummy & Tensor	Failure Mode	Lesson Taught
Drill 1: Aim & Gate	Brittle Dummy (60 HP) [σ_x=80, σ_y=0, τ_{xy}=0]	Cyan • Rankine Opt θ_p = 0.0°	Wait-for-zombie visibility gate, then turn [A]/[D] from 45° back to 0° (2θ = 0°) until gun turns green **ARMED**.
Drill 2: Shear 2θ	Sheared Dummy (80 HP) [σ_x=80, σ_y=20, τ_{xy}=30]	Cyan • Rankine Opt θ_p = 22.5°	Non-zero shear tilts the principal failure plane! Students verify physical 22.5° equals Mohr circle 2θ = 45°.
Drill 3: Ductile τ_{max}	Ductile Armor (90 HP) [σ_x=80, σ_y=20, τ_{xy}=30]	Orange • Tresca Opt θ_s = 67.5°	Identical tensor as Drill 2, but Ductile armor resists tensile σ_1. Player pivots 45° on crank (90° on Mohr) to peak shear τ_{max}.
Drill 4: Hydro [5]	Quench Target (100 HP) [σ_x=90, σ_y=30, τ_{xy}=45]	Orange -> [5] Hydro Quenched θ_p = 0.0°	Pressing [5] Hydro-Jet blasts hydrostatic pressure, setting τ_{xy} = 0 and turning orange armor into cyan brittle glass.

8. How the Game Runs Frame-by-Frame (60 FPS Main Loop)

Every browser animation frame (~16.67 ms), `mainLoop()` executes `GameEngine.update()` followed by `GameEngine.draw()`:

Step	Engine Routine	Exact Execution Logic
1. Pause Guard	`update()`	Aborts physics immediately if state ≠ playing or if any modal overlay (Welcome, Map, Guide, Lab, Report, Game Over) is open.
2. Time & FX	`slowmo / shake`	Decays screenShake (×0.88/frame); sets timeSlowFactor = 0.2 if Slow-Mo [1] is active; decrements active power-up timers.
3. Wave Spawn	`spawnNextZombie()`	Releases queued enemies from x ≈ 848–904 px (interval 55 frames if arena empty, 105 frames between staggered spawns; tutorial waits cleanly).
4. Zombie Tick	`z.update(factor)`	Advances left at speed × factor (0.5× if hydro-soaked); ticks Shapeshifter 240-frame phase flip; checks Boss phase thresholds (66%, 33% HP).
5. Barricade Check	`x <= 130`	If Shield [4] active: knocks attacker back to x = 240 with cyan sparks. Otherwise: barricade loses 15% HP, resets streak, and checks Game Over (HP ≤ 0).
6. Radar Sync	`getPrimaryTarget()`	Selects closest visible zombie (x ≤ width − inset) and updates TacticalHUD dial arc, Mohr's circle (2θ), and transformation graph live.
7. Burst Queue	`isFiringHeld / projectiles`	While [SPACE] is held & crank remains armed, auto-fires bursts on cooldown; integrates tracer velocities (44 px/f) and bouncing brass casings (g = +0.36 px/f²).
8. Clear Check	`levelCleared()`	When zombies.length === 0 and queue empty: saves stars/high score, unlocks next sector, and opens the Every-Zombie Forensic Report modal.
9. Layered Render	`draw()`	Blits cached offscreen horror scenery (<0.1 ms), draws mist/bats/rain/lamps, renders steel barricade & turret breech wheel, zombies, aim reticle, tracers, and HUD.

9. Complete Player Controls & Inverse Mohr Aiming

Input Method	Action & Granularity	Mathematical & UX Effect
[A] / [←] & [D] / [→]	Rotate Crank ±2° (Hold Shift: ±10°)	Changes physical plane orientation θ in [0°, 180°); counter-clockwise +θ follows standard Mechanics convention.
Rotary Crank Dial / Slider / Snap	Drag 360° Wheel or Snap (0°, 22.5°, 45°, 67.5°, 90°, 135°, ±1°, ±5°)	Instant keyboard-free alignment on desktop or touch screens; slider synchronizes at 0.5° resolution.
Click/Drag Mohr's Circle	Direct Interactive Radar Aiming	Computes pointer angle φ_P = atan2(dy, dx) around Center C, references φ_X = atan2(τ_{xy}, (σ_x−σ_y)/2), and solves θ = ((φ_X − φ_P)·180°)/(2π) mod 180°.
[SPACE] or FIRE GUN	Trigger / Hold Auto-Burst	Requests burst from Heavy Machine Gun; fires 4 staggered tracer rounds if zombie visible & aligned (|Δθ| ≤ 7.5°), else clicks trigger safety lock.
Keys [1] through [5]	Tactical Engineering Power-Ups	Slow-Mo [1], Hint Arc [2], Radar Scan [3], I_1 Shield [4], Hydro-Jet Quench [5] (detailed on Page 4).

10. Gun Visibility Gate, Mechanical Interlock & Machine Gun Ballistics

Version **v4.7** replaces arcade lasers and cluttered difficulty presets (Easy/Medium/Hard/Nightmare removed) with authentic, unified physics-gated heavy machine gun mechanics:

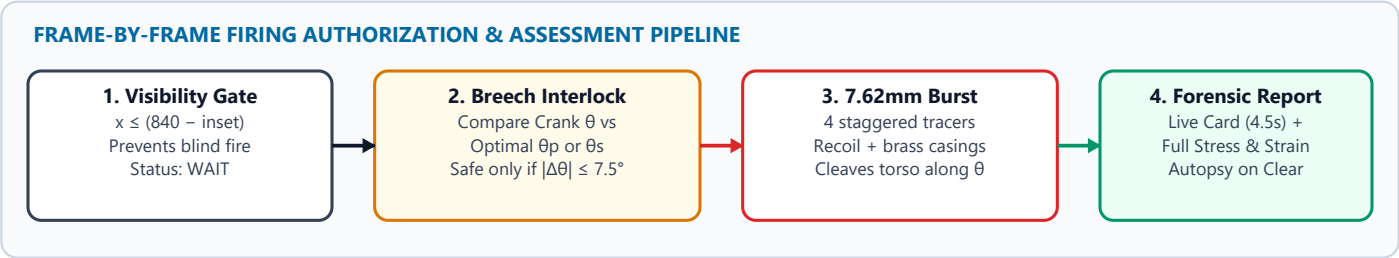


Figure 2: Strict 4-stage firing pipeline ensuring every fired bullet corresponds to an aligned solid mechanics solution.

• **Wait-for-Zombie Visibility Gate (`isZombieVisibleOnScreen`):** Spawned zombies begin just outside the right border (x ≥ 848 px on an 840 px canvas). Until an enemy's body enters `x ≤ width - max(18, round(0.34 * z.width))`, the turret tag reads **WAIT**, the firing button disables firing with an audible trigger lock click, and no ammo or bursts occur.

• **Turret Breech Handling Crank & Safety Interlock (`drawTurretAndGun`):** A circular machined crank wheel on the 7.62mm receiver rotates live with `dialAngle`, carrying a dotted green optimal plane line whenever a zombie is visible. Using `StressMechanics.evaluateAttack()`, angular error `angDiff = angleDiff180(θ, θ_opt)` governs authorization:

1. **LOCKED (angDiff > 7.5°, or > 8.5° on Titan Boss):** Receiver LED glows crimson, handle jitters, trigger clicks harmlessly, streak resets on manual pull, and **0 damage** is dealt (firing aborts cleanly before any round leaves the chamber).
2. **ARMED (angDiff ≤ 7.5°):** Reticle, crank grip, and LED blaze green. Fires a 4-round 7.62mm tracer burst (velocity 44 px/frame, 2-frame stagger) with multi-cone muzzle flash, 6.5–9 px barrel recoil, and spinning brass shell casings bouncing off the asphalt.
3. **CRITICAL HIT (angDiff ≤ 2.5°, or ≤ 3.0° on Boss):** Applies a **1.5× damage multiplier** (up to ~210 dmg), golden tracer trails, heavier camera shake, and a **3.0× accuracy score multiplier** scaled further by streak bonus `× (1 + 0.1·streak)`. On fatal hit, the zombie's torso physically separates and slides along cut angle θ!

11. Tactical Power-Ups [1–5] & Continuum Mechanics Principles

Key & Name	Charges / Timer	Battlefield Effect	Physical Solid Mechanics Rationale
[1] Slow-Mo	2 per level 300 frames (5s)	Scales enemy speed, phase timers & animation to 20% speed.	Gives students quiet calculation time to evaluate $\tan(2\theta_p) = 2\tau_{xy} / (\sigma_x - \sigma_y)$ or inspect the curve chart.
[2] Hint Arc	3 per level 240 frames (4s)	Highlights best angle arc & triggers tactical toast guidance.	Connects numerical optimal root θ_p or θ_s visually back onto the physical 180° cutting protractor.
[3] Scanner	2 per level 360 frames (6s)	Triggers radar diagnostic sweep across active targets.	Refreshes full tensor inspection across the stress element and Mohr's Circle plot.
[4] I_1 Shield	1 per level 300 frames (5s)	Projects curved cyan energy barrier at x = 145 px; bounces attackers back to x = 240 px.	Demonstrates the **First Stress Invariant**: $I_1 = \sigma_x + \sigma_y = \sigma_1 + \sigma_2 =$ Constant regardless of rotation.
[5] Hydro-Jet	2 per level 60f stream / 360f soak	Fires high-pressure water torrent hitting visible zombies; pushes back +80 px & halves walk speed for 6s.	**Hydrostatic Quenching:** Omnidirectional fluid pressure wipes deviatoric shear ($\boldsymbol{\tau_{xy} \to 0}$), converting Ductile & Shapeshifter targets into Brittle Cyan (Mohr's circle shrinks toward isotropic state; principal plane aligns with x/y axes)!

12. Analytical Solid Mechanics Engine & Vector Mohr's Circle

Every gameplay calculation, HUD vector, transformation graph, strain readout, and kill report is driven deterministically by `StressMechanics` (`js/mechanics.js`). Figure 3 illustrates how a physical plane rotation of **θ_p = 22.5° counter-clockwise** corresponds to sweeping **$2\theta_p$ = 45° clockwise** from reference face X($\sigma_{x'}$ +τ_{xy}) onto principal stress σ_1 on Mohr's Circle:

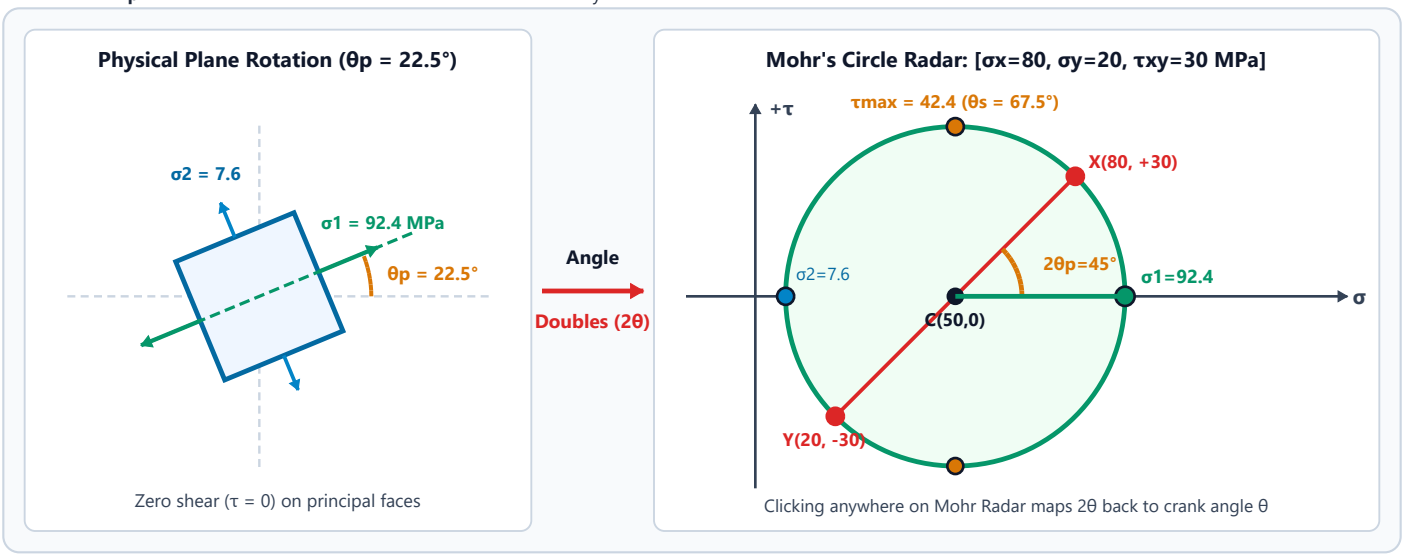


Figure 3: Vector correspondence between Physical Element rotation θp = 22.5° CCW (left) and Mohr's Circle angle 2θp = 45° CW (right) for [80, 20, 30] MPa.

13. Complete Suite of Implemented Mechanics Equations (14 Core Relations)

# & Quantity	Implemented Formula (js/mechanics.js)	Game / Physical Significance
1. Normal Stress σ_θ	$\sigma_\theta = (\sigma_x+\sigma_y)/2 + ((\sigma_x-\sigma_y)/2)\cdot\cos(2\theta) + \tau_{xy}\cdot\sin(2\theta)$	Tensile/compressive stress driving Rankine brittle cleavage.
2. Shear Stress τ_θ	$\tau_\theta = -((\sigma_x-\sigma_y)/2)\cdot\sin(2\theta) + \tau_{xy}\cdot\cos(2\theta)$	In-plane sliding traction driving Tresca ductile slip.
3. Orthogonal Stress	$\sigma_{90} = \sigma_x + \sigma_y - \sigma_\theta$, $\tau_{90} = -\tau_\theta$	Complementary stress on perpendicular face θ + 90°.
4. Mohr Center C	$C = \sigma_{hydro} = (\sigma_x + \sigma_y) / 2$	Mean / hydrostatic center along horizontal σ-axis.
5. Mohr Radius R	$R = \tau_{max} = \sqrt{[\ ((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2\ ]}$	Equals maximum in-plane shear stress and circle radius.
6. Principal Stresses	$\sigma_1 = C + R$ (at θ_p), $\sigma_2 = C - R$ (at $\theta_p + 90°$)	Extremal normal stresses with zero shear traction (τ = 0).
7. Principal & Shear θ	$\theta_p = ½\ \text{atan2}(2\tau_{xy}, \sigma_x-\sigma_y)$ (mod 180°); $\theta_s = \theta_p \pm 45°$	Checked directly via transform(θ_{p1}) so σ_1 vs σ_2 never swaps!
8. Stress Invariants	$I_1 = \sigma_x + \sigma_y = \sigma_1 + \sigma_2$; $I_2 = \sigma_x\sigma_y - \tau_{xy}^2 = \sigma_1\sigma_2$	Rotationally invariant scalars; I_1 powers Shield [4].
9. von Mises Stress	$\sigma_{vm} = \sqrt{(\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2)} = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$	Distortion-energy yield metric governing Wave 9 Titan Boss.
10. Hooke's Law (3D)	$\varepsilon_x = (\sigma_x-\nu\sigma_y)/E$, $\varepsilon_y = (\sigma_y-\nu\sigma_x)/E$, $\varepsilon_z = -\nu(\sigma_x+\sigma_y)/E$	Evaluated with steel alloy defaults E = 200 GPa, ν = 0.30 (in με).
11. Shear Moduli & γ	$G = E / [2(1+\nu)]$, $K = E / [3(1-2\nu)]$, $\gamma_{xy} = \tau_{xy} / G$	Computes shear modulus (76.92 GPa) and engineering strain (μrad).
12. Principal Strains	$\varepsilon_{1,2} = (\varepsilon_x+\varepsilon_y)/2 \pm \sqrt{[((\varepsilon_x-\varepsilon_y)/2)^2 + (\gamma_{xy}/2)^2]}$, $\gamma_{max} = 2R_\varepsilon$	Plotted dynamically on post-level Strain Autopsy matrices.
13. Strain Transform	$\varepsilon_\theta = avg + diff\cdot\cos(2\theta) + (\gamma_{xy}/2)\cdot\sin(2\theta)$; $\gamma_\theta = -2diff\cdot\sin(2\theta) + \gamma_{xy}\cdot\cos(2\theta)$	Drives the green dashed deformed element in Study/Report Labs.
14. Dilatation & Energy	$e = \varepsilon_x+\varepsilon_y+\varepsilon_z$; $u = ½(\sigma_x\varepsilon_x+\sigma_y\varepsilon_y+\tau_{xy}\gamma_{xy})$; $u_d = [(1+\nu)/(3E)]\sigma_{vm}^2$	Partitions total strain energy u (kJ/m³) into volume change u_v & shape change u_d.

14. Worked Numerical Benchmark (Standard Test Specimen: [80, 20, 30] MPa)

Both the live game engine (**StudyLab** Preset #1 & Level 4 Dragger) and **test_mechanics.py** benchmark against this exact plane stress state (E = 200 GPa, ν = 0.30, G = 76,923.08 MPa):

Calculation Step	Analytical Substitution	Computed Engine Output
Mohr Center & Radius	$C = (80 + 20) / 2 = 50.0$ MPa; $R = \sqrt{[(80 - 20)/2)^2 + 30^2]} = \sqrt{(900 + 900)}$	**$C = 50.0$ MPa, $R = \tau_{max} = 42.43$ MPa**
Principal Stresses	$\sigma_1 = 50 + 42.426$; $\sigma_2 = 50 - 42.426$	**$\sigma_1 = 92.43$ MPa, $\sigma_2 = 7.57$ MPa**
Optimal Angles θ_p, θ_s	$\tan(2\theta_p) = 2(30)/(80-20) = 1.0$ -> $2\theta_p = 45°$; $\theta_s = \theta_p + 45°$	**$\theta_p = 22.5°$ (Brittle), $\theta_s = 67.5°$ (Ductile)**
Aligned Cut (θ = 22.5°)	$\sigma_{22.5} = 50 + 30\cos(45°) + 30\sin(45°)$; $\tau_{22.5} = -30\sin(45°) + 30\cos(45°)$	**$\sigma_\theta = 92.4$ MPa, $\tau_\theta = 0.0$ MPa (CRIT CLEAVE!)**
Misaligned Cut (θ = 90°)	$\sigma_{90} = 50 + 30\cos(180°) + 30\sin(180°)$; $\tau_{90} = -30\sin(180°) + 30\cos(180°)$	**$\sigma_\theta = 20.0$ MPa, $\tau_\theta = -30.0$ MPa (LOCKED / 0 DMG)**
Ductile Shear (θ = 67.5°)	$\tau_{67.5} = -30\sin(135°) + 30\cos(135°) = -21.21 - 21.21$	**$\lvert\tau_\theta\rvert = 42.4$ MPa (TRESCA SHEAR TEAR)**
von Mises & Invariant I_1	$I_1 = 80 + 20 = \sigma_\theta + \sigma_{\theta+90}$; $\sigma_{vm} = \sqrt{(92.43^2 - 92.43 \cdot 7.57 + 7.57^2)}$	**$I_1 = 100.0$ MPa, $\sigma_{vm} = 88.88$ MPa**
3D Hooke Strains	$\varepsilon_x = (80 - 0.3 \cdot 20)/0.2$; $\varepsilon_y = (20 - 0.3 \cdot 80)/0.2$; $\varepsilon_z = -0.3(100)/0.2$; $\gamma_{xy} = 30/76.923$	**ε_x=370 με, ε_y=−20 με, ε_z=−150 με, γ_{xy}=390 μrad**
Principal Strain @ 22.5°	$\varepsilon_{22.5} = 175 + 195\cos(45°) + 195\sin(45°)$; $\gamma_{22.5} = 0$	**$\varepsilon_1 = 450.8$ με, $\gamma_\theta = 0.0$ μrad**
Dilatation & Energies	$e = 370 - 20 - 150$; $u = \frac{1}{2}(80 \cdot 370 + 20 \cdot (-20) + 30 \cdot 390) \times 10^{-3}$	**$e = 200.0$ με, $u = 20.45$ kJ/m³ ($u_d = 17.12$ = 83.7%)**

15. Automated Verification & Structural Code Audit Results

Both mathematical fidelity and DOM/code integrity are verified automatically via unit tests and static analysis scripts:

Verification Test	Target Module & Property Verified	Observed Result
1. First Stress Invariant (I_1)	Verifies $I_1 = \sigma_\theta + \sigma_{\theta+90} = 100.0$ MPa across all 180°	**PASS**
2. Mohr Geometry (C, R)	Matches Center = 50.0 MPa and Radius = 42.43 MPa	**PASS**
3. Principal Stresses (σ_1, σ_2)	Matches $\sigma_1 = 92.4$ MPa, $\sigma_2 = 7.6$ MPa	**PASS**
4. Principal & Shear Angles	Matches $\theta_p = 22.5°$ and $\theta_s = 67.5°$	**PASS**
5. Optimal Hit Transformation	$\theta = 22.5°$ yields $\sigma_\theta = 92.4$ MPa and $\tau_\theta = 0.0$ MPa	**PASS**
6. Suboptimal Deflection	$\theta = 90.0°$ yields $\sigma_\theta = 20.0$ MPa, $\tau_\theta = -30.0$ MPa	**PASS**
7. Tresca Shear Condition	$\theta = 67.5°$ yields $\lvert\tau_\theta\rvert = 42.4$ MPa (τ_{max})	**PASS**
8. von Mises Criterion	Tensor form matches principal formula: $\sigma_{vm} = 88.9$ MPa	**PASS**
9. Generalized Hooke's Law	ε_x=370 με, ε_y=−20 με, ε_z=−150 με, γ_{xy}=390 μrad	**PASS**
10. Strain Transformation	$\varepsilon_\theta = 450.8$ με max & $\gamma_\theta = 0.0$ μrad at $\theta_p = 22.5°$	**PASS**
11. Dilatation & Energy Density	$e = 200.0$ με, $u = 20.45$ kJ/m³, $u_d = 17.12$ kJ/m³ (0.001s runtime)	**PASS (11/11)**
12. Static DOM & Server Probe	148 DOM IDs (0 missing), balanced JS/CSS syntax, HTTP 200 on all 8 v4.7 routes	**PASS (24/24)**

16. Zombie Bestiary & Classical Material Failure Mapping

Enemy Class & Visuals	Failure Law & Formula	Combat Mechanics, Tolerances & HP
1. Brittle Biter Sage skin, brown coat, red striped tie, exposed shin bone, bugging eyes, Cyan glow	**Rankine Theory** (Max Normal Stress) $\sigma_1 \geq \sigma_Y$	Fails exclusively by tensile cleavage at $\theta = \theta_p$ (where shear vanishes, $\tau = 0$). Tolerance: ±7.5° (Crit ≤ 2.5°). Base Damage: 110–140 (+50% Crit).
2. Ductile Dragger Burnt-orange survivor coat, molten amber eyes, Orange aura	**Tresca Criterion** (Max Shear Stress) $\tau_{max} \geq \sigma_Y/2$	Resists normal cuts! Yields along 45°-shifted slip planes $\theta_s = \theta_p \pm 45°$. Can also be Hydro-Quenched [5] ($\tau_{xy} \to 0$) into Brittle Cyan.
3. Quantum Shapeshifter Ashen skull, teal/magenta coat, crimson hollow eyes + drool	**Dynamic Rankine ↔ Tresca** Alternating Tensor State	Every **240 frames** (~4s), mutates random [σ_x, σ_y, τ_{xy}] and swaps phase between Tensile (Cyan/Violet) and Shear (Pink/Orange). Forces rapid crank re-aiming.
4. Linked Twins Teal street pair, huge yellow bulb cartoon eyes, Green aura	**Coupled Plane Cleave** $\theta = \theta_p$ (Twin Pair)	Walks offset vertically sharing identical [σ_x, σ_y, τ_{xy}] and θ_p. Player locks once on primary twin and bursts both rapidly.
5. von Mises Titan Boss Colossal 95×145 px brute, purple-crimson coat, Red aura (600 HP)	**Distortion Energy** $\sqrt{(\sigma_\theta^2 + 3\tau_\theta^2)} \geq \sigma_Y$	**3-Phase Boss** (±8.5° tolerance, 85 dmg/burst): shifts to [90, 30, −45] @ ≤ 66% HP (1.15× spd) & [110, 45, 55] @ ≤ 33% HP (1.35× spd).

17. Complete 10-Sector Campaign Progression & Horror Scenery Engine

Level progression is gated sequentially via `maxUnlockedLevel` and graded **1 to 3 Stars** (3 Stars ≥ 80% crit accuracy, 2 Stars ≥ 50%, 1 Star < 50%). Scenery is pre-rendered onto an offscreen canvas cache (<0.1 ms/frame blit) plus live animated mist, flickering streetlamps, bats, puddles, and embers:

Level & Theme	Spawned Wave Enemies & [σ_x, σ_y, τ_{xy}] MPa	Mechanics Learning Objective
L1: Straight Pull *Cemetery Night*	3 Brittle (90–110 HP): [80,0,0], [100,0,0], [120,0,0]	Pure uniaxial tension (τ_{xy}=0); target aligns flat at θ = 0°/90°.
L2: Two-Way Pull *Cemetery Night*	3 Brittle (110–120 HP): [90,30,0], [110,40,0], [80,20,0]	Biaxial loading shifts Mohr center C = (σ_x+σ_y)/2 without rotating axes.
L3: Angled Shear *Cemetery Night*	3 Brittle (120–140 HP): [80,20,30], [60,20,40], [90,10,−35]	Shear tilts principal axes; teaches 2θ Mohr radar rotation & sign (±τ_{xy}).
L4: Brittle & Ductile *Cursed Woods*	2 Brittle + 2 Ductile: [80,30,25], [80,20,30], [70,10,40], [95,25,−30]	Distinguishes Cyan σ_1 cleavage (θ_p) vs. Orange τ_{max} shear slip ($\theta_s = \theta_p \pm 45°$).
L5: Armored Wave *Cursed Woods*	3 Ductile + 1 Brittle (140–160 HP): [90,30,45], [85,25,40], [75,20,35], [100,30,−25]	Trains [5] Hydro-Jet quenching to collapse deviatoric shear into brittle glass.
L6: Color Changers *Cursed Woods*	2 Shapeshifters + 1 Brittle (130–160 HP): [75,25,35], [85,15,20], [70,20,−30]	Adapts crank ±45° dynamically when tensor mutates every 240 frames.
L7: Linked Twins *Blood Overpass*	4 Coupled Twins (120 HP pairs): [85,15,30] ×2 & [75,25,−25] ×2	Identifies shared stress state on marching pairs for rapid double cleaves.
L8: Speed Rush *Blood Overpass*	Fast Swarm (0.65–0.72 px/f): Brittle [110,40,35], Ductile [80,20,35], Shift [85,25,−30]	Combines [4] I_1 Shield kinetic barrier & rapid targeting under speed pressure.
L9: Titan Boss *Blood Overpass*	VON MISES TITAN COLOSSUS (600 HP): Starts [110,25,45], mutates 3 phases	Sustained burst fire tracking distortion energy & mid-battle phase shifts.
L10+: Endless Mode *Blood Overpass*	4 + level randomized mixed enemies; scaling speed & HP (120 + 15·lvl)	Unlimited endurance survival leaderboard after conquering Campaign L1–L9.

18. Three-Tier Continuous Assessment & Every-Zombie Report System

Unlike games that hide physics after combat ends, every single eliminated zombie produces immediate telemetry AND a full forensic record:

Assessment Layer	UI Component / Class	Exact Data & Interactive Controls Provided
1. Live Non-Blocking Kill Card (In-Combat)	`#liveKillCard` `showLiveKillReport()`	Appears top-right for 4.5s on EVERY kill without pausing waves. Prints Kill #, Score (+CRIT badge), Cut angle θ, Mohr 2θ, $\sigma_{\theta'}$ $\tau_{\theta'}$, failure criterion title, and exact optimal angle θ_{opt}.
2. Post-Level Every-Zombie Forensic Inspector	`#reportModal` + `#reportLabWrapper`	**Always visible** on level clear (and linked from Game Over whenever kills > 0): includes Summary Grid (Accuracy, Crits, Misfires, Streak), **All-Slain Zombie Kill Report Table** with per-row *VIEW* buttons, specimen selector tabs, Exploration Slider θ (0°–180° + *MY CUT* reset), dual Mohr + Deformed Element canvases, Energy Partition bar (u_v vs u_d), and 6-step derivations.
3. Standalone Study Lab & Codex Sandbox	`#studyModal` `class StudyLab`	Non-combat faculty/student laboratory with free sliders for σ_x, σ_y, τ_{xy} (−100..100 MPa) & θ, 4 classic presets (Worked Example #1, Pure Shear, Uniaxial, Hydrostatic), and import buttons for recently defeated zombies.

19. Procedural Web Audio Architecture (Zero Audio Assets)

All sound effects and adaptive music in `SoundController` (`js/audio.js`, 1,239 lines) are synthesized mathematically in the browser via the Web Audio API (oscillators, filtered noise buffers, wave-shaper distortion, feedback echoes):

Audio Subsystem	Synthesis Architecture & Behavioral Triggers
7.62mm Machine Gun Bursts & Locks	4 staggered transient noise cracks (0.044s gap) + low-end punch + brass shell ping on valid burst; mechanical dry trigger click (`playTriggerLock`) when fired off-angle or before zombie enters screen.
Tactical Proximity & Power-Up SFX	Rotary crank emits pitch-modulated ticks proportional to alignment proximity; distinct Hydro-Jet water deluge wash, Invariant Shield chime, Boss alarm siren, and streak combo fanfares.
Dynamic Dual-Mode Horror BGM & Roar	Real-time step sequencer switching smoothly between **Ambient** (74 BPM heartbeat/clock dread on menus) and **Combat** (84–96 BPM sub-bass & stings), plus a 4-layer Welcome Zombie Roar synthesizer.

20. How to Run, Test & Share with Faculty or Students

1. Local Instant Run (Recommended for Demo): Execute `python run_game.py` in the project folder. It starts a no-cache Python HTTP server on port 8000, auto-opens **http://localhost:8000/** in Chrome, and prints the shareable local Wi-Fi URL (`http://<LAN-IP>:8000/`) for classmates on the same network.
2. Run Automated Mechanics Verification: Execute `python test_mechanics.py` to verify all **11/11** stress/strain/energy calculations cleanly.
3. Public Chrome Link (Permanent Cloud Hosting): Because the project is 100% static client-side HTML/CSS/JS with relative paths, uploading the folder to **Netlify Drop** or **GitHub Pages** produces an instant HTTPS URL playable globally in any browser.